

Revised Generation Interconnection System Impact Study Report

For Queue Project Q18-1073

Lincoln 230 kV
October 21, 2019

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q18-1073. The Interconnection Customer proposes a 93.4 MW battery energy storage facility interconnecting at the Lincoln 230 kV substation in the Illinois area. The request was received on October 08, 2018. The study evaluated system intact and single contingency conditions using the regional planning case, including 430 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 25 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$3,958,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
IMPERIAL - GRUVER 230kV line	IMPERIAL - EUFAULA 500kV	129.5	720	6.6
LOOKINGGLASS - CORRIDOR 230kV	TREGO SUBSTATION - BEAVER 500kV line	128.9	1,160	9.9
LOOKINGGLASS - TRUMAN 230kV	BEAVER - TREGO 500kV	127.7	1,195	9.9
MOUTON - LEDGES 230kV	EL PASO - RAVENNA SUBSTATION 500kV Line	127.4	675	5.8
CANAL ROAD - NEWPORT TIE 230kV	BEAVER - TREGO 500kV	127.2	1,484	10.4
TRUMAN - WALDO 230kV line	TREGO - BEAVER 500kV Line	126.7	1,225	9.9
EUFAULA - JOSO 230kV Line	BEAVER - EUFAULA 500kV	125.0	987	8.1
INVOLTA SUBSTATION - OLD FORGE 230kV L	RAVENNA - HYLEBOS 500kV	124.9	520	5.3
WINTHROP - HEBO 230kV Line	HEBO - BARRETT 230kV	124.7	993	5.5
GRAY'S FLAT - E METHUEN 230kV	E METHUEN - MADLINE 230kV L	124.0	1,521	8.9
ADIRONDACK - JAMES DE YOUNG 230kV	ST. LAWRENCE - JAMES DE YOUNG 230kV	123.4	1,282	7.4
MUSKOE - DARLINGTON RD 100kV L	DEKALB - RICHLAND 500kV	123.0	515	6.7
TREAT - MOUTON SUBSTATION 230kV	EL PASO - RAVENNA 500kV	121.6	722	5.8
JOSO - ONANCOCK DIESEL GENERATION FACILITY 230kV	BEAVER - EUFAULA 500kV	121.2	2,332	14.3
BURRIS PIT - HEBO 230kV	TREGO - BEAVER SUBSTATION 500kV	120.7	1,482	7.9

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
E METHUEN 500/230kV Transformer Bank	BEAVER - TREGO 500kV	120.3	2,846	16.0
MAPLE ST - LEETONIA 230kV	EL PASO - RAVENNA 500kV	119.8	829	5.2
NEWPORT TIE - POST ROCK 230kV Line	TREGO SUBSTATION - BEAVER 500kV	119.4	2,343	13.2
ATASCOCITA - MAPLE STREET 230kV	EL PASO - RAVENNA 500kV	119.2	848	5.2
GALION GENERATING STATION 500 kV/230 kV Transformer	TREGO - BEAVER SUBSTATION 500kV	119.1	1,792	10.8
ADAMS - ONANCOCK DIESEL GENERATION FACILITY 230kV	TREGO - BEAVER 500kV line	118.0	2,802	15.9
ADAMS - PICKFORD SUBSTATION 230kV L	BEAVER - TREGO 500kV	117.8	1,450	8.1
MYRTLE PT - POST ROCK 230kV Line	TREGO - BEAVER 500kV	117.7	2,391	12.8
RAVENNA - OLD FORGE 230kV	RAVENNA - HYLEBOS 500kV	117.4	1,367	8.4
ADIRNDACK 500/230 kV XFMR	E METHUEN 500/230 kV XFMR	117.0	1,570	5.1

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
GRUVER - IMPERIAL 230kV L	Rebuild line with higher-capacity conductor	\$7,221,000	1% (\$47,000)
LOOKINGGLASS - CORRIDOR 230kV Line	Rebuild line with higher-capacity conductor	\$25,610,000	1% (\$165,000)
LOOKINGGLASS - TRUMAN 230kV	Rebuild line with higher-capacity conductor	\$28,362,000	1% (\$183,000)
MOUTON SUBSTATION - LEDGES 230kV	Rebuild line with higher-capacity conductor	\$8,400,000	1% (\$54,000)
CANAL ROAD - NEWPORT TIE 230kV	Rebuild line with higher-capacity conductor	\$28,261,000	1% (\$182,000)
WALDO - TRUMAN 230kV	Rebuild line with higher-capacity conductor	\$20,181,000	1% (\$130,000)
JOSO - EUFAULA 230kV	Rebuild line with higher-capacity conductor	\$10,525,000	1% (\$68,000)
OLD FORGE - INVOLTA 230kV	Rebuild line with higher-capacity conductor	\$25,935,000	1% (\$167,000)
WINTHROP - HEBO 230kV	Rebuild line with higher-capacity conductor	\$13,504,000	1% (\$87,000)
GRAY'S FLAT - E MTHUEN 230kV	Rebuild line with higher-capacity conductor	\$36,690,000	1% (\$236,000)

Facility	Required Upgrade	Cost Estimate	Allocation to Project
ADIRONDACK - JAMES DE YOUNG 230kV	Rebuild line with higher-capacity conductor	\$26,858,000	1% (\$173,000)
DARLINGTON RD - MUSKOGEE 100kV	Rebuild line with higher-capacity conductor	\$13,899,000	1% (\$89,000)
MOUTON - TREAT 230kV	Rebuild line with higher-capacity conductor	\$22,301,000	1% (\$143,000)
JOSO - ONANCOCK DIESEL GENERATION FACILITY 230kV line	Rebuild line with higher-capacity conductor	\$23,079,000	1% (\$149,000)
BURRIS PIT - HEBO 230kV	Rebuild line with higher-capacity conductor	\$15,612,000	1% (\$100,000)
EAST METHUEN 500/230 kV Autotransformer	Replace transformer with higher-rated unit	\$37,823,000	1% (\$244,000)
MAPLE ST - LEETONIA 230kV	Rebuild line with higher-capacity conductor	\$27,704,000	1% (\$178,000)
NEWPORT TIE - POST ROCK 230kV Line	Rebuild line with higher-capacity conductor	\$26,796,000	1% (\$173,000)
ATASCOCITA - MAPLE STREET 230kV	Rebuild line with higher-capacity conductor	\$39,796,000	1% (\$255,000)
GALION GENERATING STA 500-230 kV XFMR	Replace transformer with higher-rated unit	\$35,582,000	1% (\$230,000)
ADAMS - ONANCOCK DIESEL GENERATION FACILITY 230kV line	Rebuild line with higher-capacity conductor	\$49,682,000	1% (\$321,000)
ADAMS - PICKFORD SUBSTATION 230kV L	Rebuild line with higher-capacity conductor	\$24,070,000	1% (\$155,000)
POST ROCK - MYRTLE POINT SUBSTATION 230kV	Rebuild line with higher-capacity conductor	\$21,581,000	1% (\$139,000)
RAVENNA - OLD FORGE 230kV	Rebuild line with higher-capacity conductor	\$18,711,000	1% (\$120,000)
ADIRNDACK 500/230 kV XFMR	Replace transformer with higher-rated unit	\$26,525,000	1% (\$170,000)
	Total allocated to project		\$3,958,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.